

# Naman Jain

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## Education

### B.E. Computer Science and Engineering (Cyber Security)

MS Ramaiah Institute of Technology, Bangalore

2023 – 2027

CGPA: 9.07 / 10

## Experience

### Hewlett Packard Enterprise (HPE) — Career Preview Program Intern

2025 - Present

- Currently working on **PUMA-Boot**, a secure UEFI-based node provisioning framework for HPC clusters aimed at improving trust and security during node bootstrapping.
- Exploring firmware-level attestation, TPM-based verification, and post-quantum cryptographic mechanisms for secure OS provisioning and node authentication.
- Contributing to the design and implementation of secure boot workflows, encrypted artifact delivery, and integrity verification mechanisms.

### Flutter Development Intern, MSRIT

Aug 2024

- Developed a responsive travel booking application UI using Flutter and Dart with modular, reusable widget architecture following Material Design guidelines.

## Projects

### XTI-SOC — Explainable AI Security Operations Framework

2026   🐙 GitHub

*Python, FastAPI, Next.js, React (TypeScript), XGBoost, SHAP, Scapy, SQLite, WebSocket*

- Built a lab-scale SOC framework integrating real-time packet capture, ML-based intrusion detection, and explainable AI — trained on 9.7M+ labeled network flows from CICIDS2018.
- Engineered a dual-pipeline XGBoost classifier (binary + 7-class multiclass) achieving 0.83 ROC-AUC with chronological train/test split; eliminated data leakage via RobustScaler fitted on training data only.
- Implemented SHAP TreeExplainer to generate per-alert feature importance, translating black-box ML decisions into analyst-readable forensic insights rendered on the dashboard.
- Built temporal correlation engine detecting Port Scans, DDoS, Brute Force, and ARP Spoofing across 60-second rolling windows; integrated AbuseIPDB CTI enrichment with L1/L2 caching.
- Designed a real-time Next.js SOC dashboard with WebSocket-streamed alerts, risk scoring ( $ML \times 0.7 + CTI \times 0.3$ ), and SQLite WAL storage — sub-100ms end-to-end latency.

### RoadIntel — AI Traffic Intelligence Platform

2025   🐙 GitHub

*React (TypeScript), Node.js, Express, MongoDB, YOLO, Gemini API, Leaflet*

- Built a geolocated traffic incident reporting platform with RESTful backend services, role-based access control, and real-time geospatial heatmap dashboards using Leaflet.
- Integrated YOLO-based computer vision for vehicle detection from video streams and Google Gemini API for AI-powered incident severity classification.
- Published research at **IEEE ICCECE 2026** International Conference.

### HealTrip — AI Medical Tourism Platform

2025   🐙 GitHub   🌐 Live Demo

*React, Node.js, Express, MongoDB Atlas, FastAPI, Scikit-learn, Razorpay, Stripe, Clerk*

- Developed a medical tourism platform connecting international patients with verified hospitals; designed MongoDB schemas and REST APIs for treatment booking and itinerary management.
- Built an ML microservice using Scikit-learn and FastAPI for hospital ranking and personalized recommendations; integrated Razorpay and Stripe for secure global payments.
- Reached **Top 12 out of 170+ teams** at WHACKIEST'25 Hackathon, MSRIT.

## Achievements

- **IEEE Publication:** *Road Intelligence System for Real-Time Hazard Detection and Navigation Assistance* — IEEE ICCECE 2026.
- **Top 12 / 170+ Teams** — WHACKIEST'25 Hackathon organized by CodeRit Club, MSRIT.
- Selected for **National Round of NCSC** for the Smart Elevator project.

## Technical Skills

**Languages:** Java, Python, JavaScript

**Full Stack:** React.js, Next.js, Node.js, Express.js, MongoDB, HTML, CSS

**Security:** Scapy, NFStream

**Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Cryptography & Network Security

**Tools:** Git, GitHub, VS Code, Antigravity